Experiment 16

Random Access File – Employee Details

code:

```
\#include <stdio.h>
#include <stdlib.h>

struct Employee {
    int empId;
    char empName[50];
    float empSalary;
};

int main() {
    FILE *fp;
    struct Employee emp;
    int choice, id;

    fp = fopen("employee.dat", "rb+");

    if (fp == NULL) {
        fp = fopen("employee.dat", "wb+");
    }

    if (fp == NULL) {
        printf("File opening failed.\n");
        return 1;
    }

    do {
        printf("\n--- EMPLOYEE DATABASE ---\n");
        printf("1. Add Employee\n");
        printf("2. Display Employee\n");
```

```
printf("3. Update Employee\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {

case 1:
    printf("Enter Employee ID (1-100): ");
    scanf("%d", &emp.empId);

    if (emp.empId < 1 || emp.empId > 100) {
        printf("Invalid Employee ID.\n");
        break;
    }

    fseek(fp, (emp.empId - 1) * sizeof(emp), SEEK_SET);

    if (fread(&emp, sizeof(emp), 1, fp) == 1 &&
        emp.empId != 0) {
        printf("Employee already exists.\n");
        break;
    }

    printf("Enter Employee Name: ");
    scanf(" %49[^\n]", emp.empName);

    printf("Enter Employee Salary: ");
    scanf("%f", &emp.empSalary);

    fseek(fp, (emp.empId - 1) * sizeof(emp), SEEK_SET);
    fwrite(&emp, sizeof(emp), 1, fp);
```

```
    printf("Employee added successfully.\n");
    break;

case 2:
    printf("Enter Employee ID to display: ");
    scanf("%d", &id);

    if (id < 1 || id > 100) {
        printf("Invalid Employee ID.\n");
        break;
    }

    fseek(fp, (id - 1) * sizeof(emp), SEEK_SET);

    if (fread(&emp, sizeof(emp), 1, fp) == 1 &&
        emp.empId == id) {

        printf("\nEmployee Details\n");
        printf("ID: %d\n", emp.empId);
        printf("Name: %s\n", emp.empName);
        printf("Salary: %.2f\n", emp.empSalary);

    } else {
        printf("Employee not found.\n");
    }
    break;

case 3:
    printf("Enter Employee ID to update: ");
    scanf("%d", &id);
```

```
    if (id < 1 || id > 100) {
        printf("Invalid Employee ID.\n");
        break;
    }

    fseek(fp, (id - 1) * sizeof(emp), SEEK_SET);

    if (fread(&emp, sizeof(emp), 1, fp) == 1 &&
        emp.empId == id) {

        printf("Enter New Employee Name: ");
        scanf(" %49[^\n]", emp.empName);

        printf("Enter New Salary: ");
        scanf("%f", &emp.empSalary);

        fseek(fp, (id - 1) * sizeof(emp), SEEK_SET);
        fwrite(&emp, sizeof(emp), 1, fp);

        printf("Employee updated successfully.\n");

    } else {
        printf("Employee not found.\n");
    }
    break;

case 4:
    printf("Exiting program.\n");
    break;

default:
    printf("Invalid choice.\n");
```

```
        }

    } while (choice != 4);

    fclose(fp);

    return 0;
}
```

Output:

